2024-2025_HW_A2_18. (1)Electric fields_electric force and electric field due to a point charge (E and V)_6_60min

NO. 1

a. By reference to electric field lines, explain why, for points outside an isolated spherical conductor, the charge on the sphere may be considered to act as a point charge at its centre.

.....
.....
.....
..... [2]

b. Two isolated protons are separated in a vacuum by a distance *x*.

i. Calculate the ratio

$$\frac{\text{electric force between the two protons}}{\text{gravitational force between the two protons}}.$$

ratio = ... [3]

ii. By reference to your answer in **(i)**, suggest why gravitational forces are not considered when calculating the force between charged particles.

.....
..... [1]

[Total: 6]

NO. 2

A metal sphere of radius R is isolated in space.
Point P is a distance x from the centre of the sphere, as illustrated in Fig. 7.1.

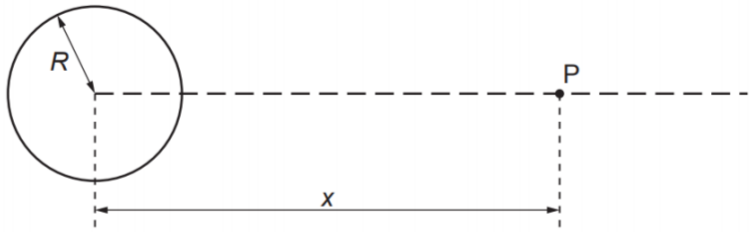

Fig. 7.1

The variation with distance x of the electric field strength E due to the charge on the sphere is shown in Fig. 7.2.

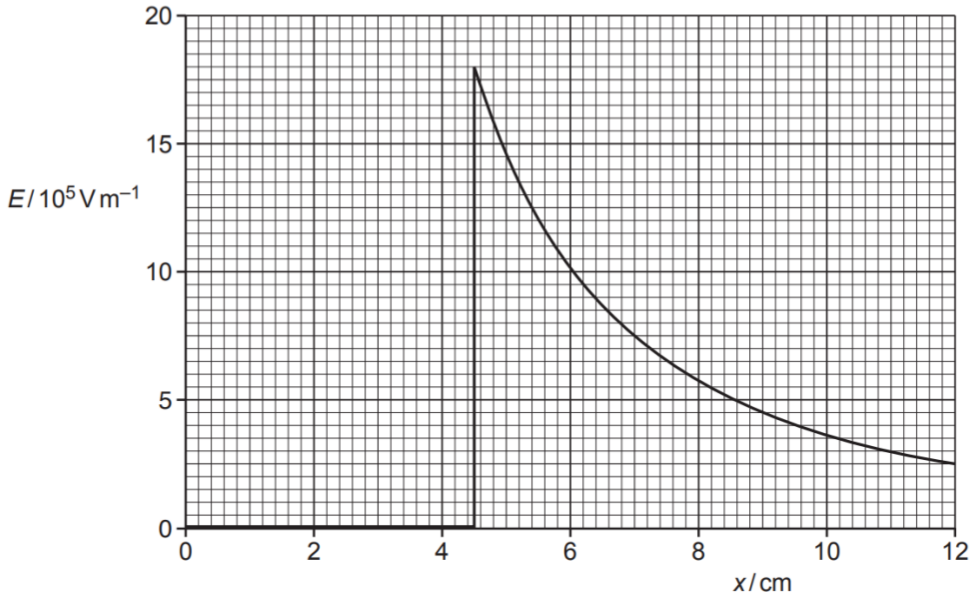

Fig. 7.2

a. State what is meant by *electric field strength*.

.....
.....
..... [2]

b. i. Use Fig. 7.2 to determine the radius R of the sphere. Explain your working.

R = ... cm [2]

ii. Use Fig. 7.2 to determine the charge Q on the sphere.

Q = ... C [3]

c. An α-particle is situated a distance 8.0cm from the centre of the sphere. Calculate the acceleration of the α-particle.

acceleration = ... ms^{-2} [3]

[Total: 10]

NO. 3

a. State what is meant by *electric field strength*.

.....
.....[1]

b. An isolated metal sphere A of radius 26cm is positively charged. Sphere A is shown in Fig. 6.1.

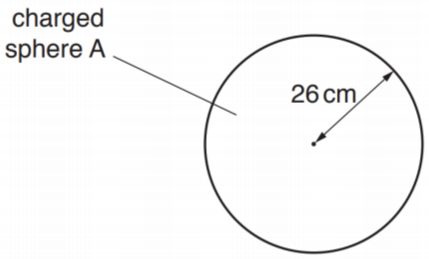

Fig. 6.1

Electrical breakdown (a spark) occurs when the electric field strength at the surface of the sphere exceeds $2.0 \times 10^4 \text{Vm}^{-1}$.
Calculate the maximum charge Q that can be stored on the sphere.

Q = ... C [2]

c. A second isolated metal sphere B, also with charge $+Q$, has a radius of 52cm.
Calculate the additional charge, in terms of Q, that may be stored on this sphere before electrical breakdown occurs.

additional charge = ...[2]

[Total: 5]

NO. 4

A solid metal sphere of radius R is isolated in space. The sphere is positively charged so that the electric potential at its surface is V_S. The electric field strength at the surface is E_S.

a. On the axes of Fig. 6.1, show the variation of the electric potential with distance x from the centre of the sphere for values of x from $x = 0$ to $x = 3R$.

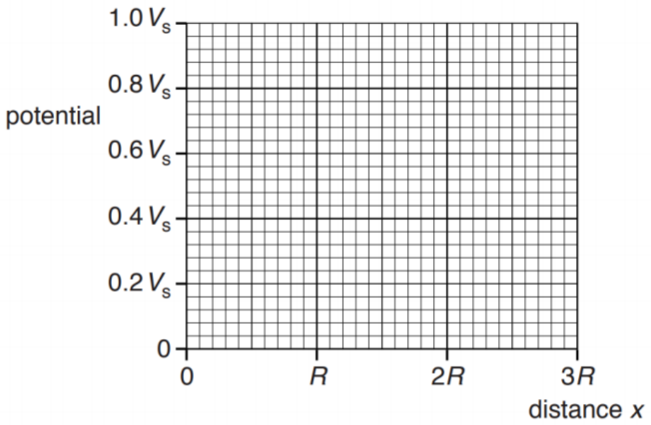

Fig. 6.1

[3]

b. On the axes of Fig. 6.2, show the variation of the electric field strength with distance x from the centre of the sphere for values of x from $x = 0$ to $x = 3R$.

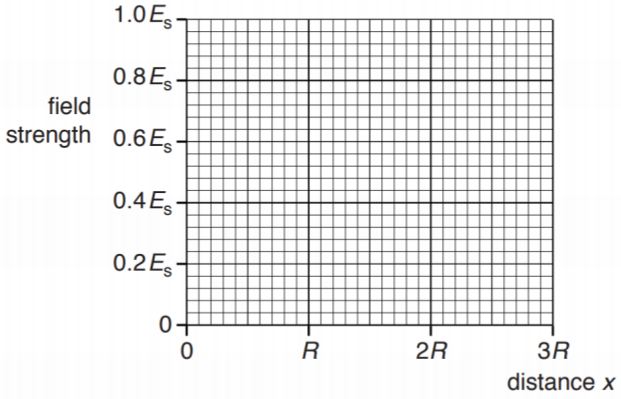

Fig. 6.2

[3]

[Total: 6]

NO. 5

a. For any point outside a spherical conductor, the charge on the sphere may be considered to act as a point charge at its centre. By reference to electric field lines, explain this.

.....
.....
.....
.....[2]

b. An isolated spherical conductor has charge q, as shown in Fig. 6.1.

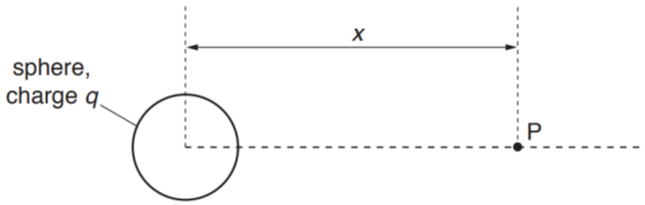

Fig. 6.1

Point P is a movable point that, at any one time, is a distance x from the centre of the sphere. The variation with distance x of the electric potential V at point P due to the charge on the sphere is shown in Fig. 6.2.

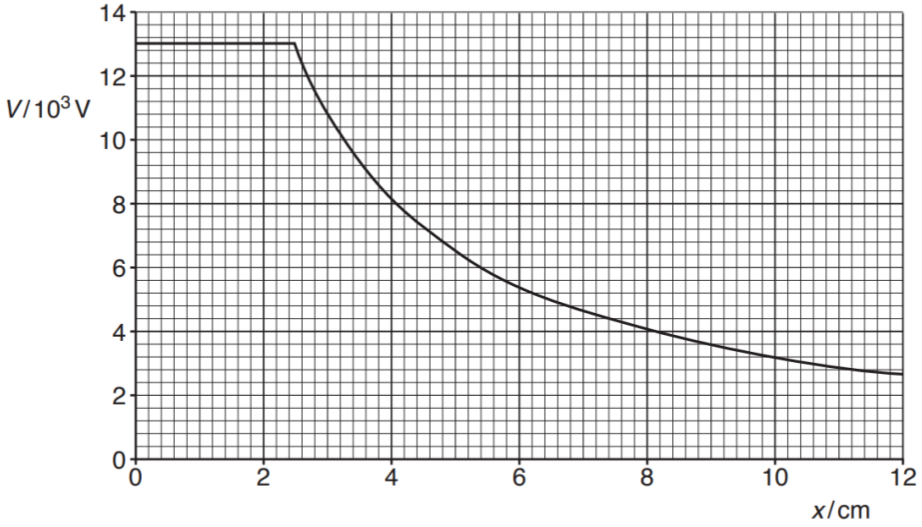

Fig. 6.2

Use Fig. 6.2 to determine

i. the electric field strength E at point P where x = 6.0cm,

E = ... NC^{-1} [3]

ii. the radius R of the sphere. Explain your answer.

R = ... cm [2]

[Total: 7]

NO. 6

A charged point mass is situated in a vacuum. A proton travels directly towards the mass, as illustrated in Fig. 4.1.

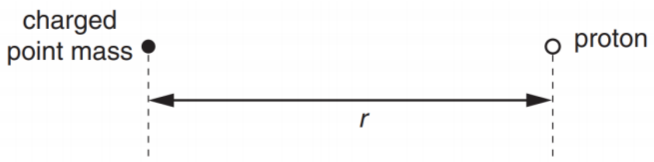

Fig. 4.1

When the separation of the mass and the proton is r, the electric potential energy of the system is U_p. The variation with r of the potential energy U_p is shown in Fig. 4.2.

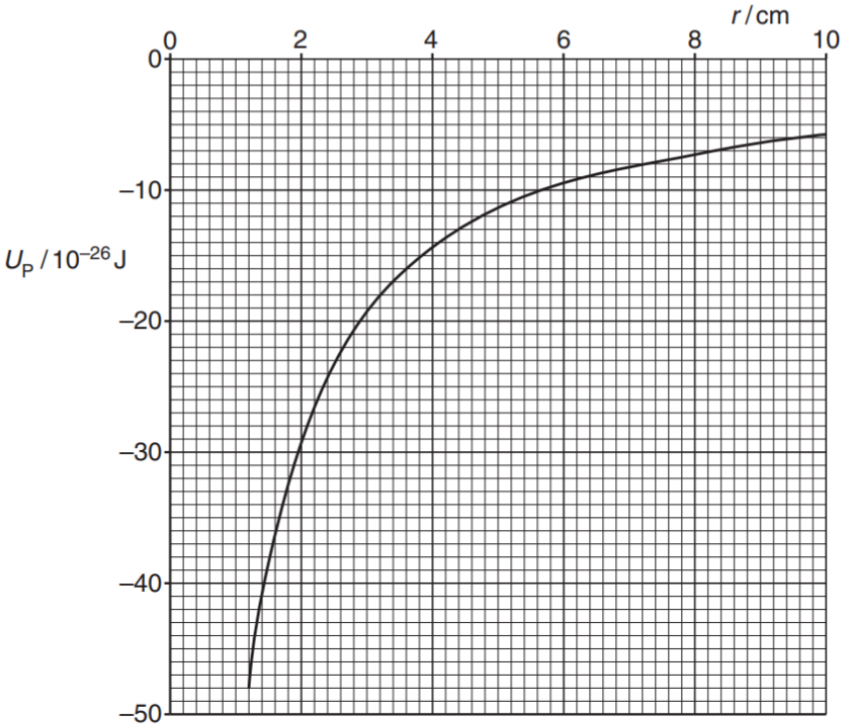

Fig. 4.2

a. i. Use Fig. 4.2 to state and explain whether the mass is charged positively or negatively.

.....
.....
.....
.....[2]

ii. The gradient at a point on the graph of Fig. 4.2 is G.

Show that the electric field strength E at this point due to the charged point mass is given by the expression

$$Eq = G$$

where q is the charge at this point.

.....
.....
.....
.....[2]

b. Use the expression in **(a)(ii)** and Fig. 4.2 to determine the electric field strength at a distance of 4.0 cm from the charged point mass.

field strength = ... V m^{-1} [4]

[Total: 8]

2024-2025_HW_A2_18. (2)Electric fields_uniform electric fields (E and V)_4_40min

NO. 1

a. Define *electric field strength*.

.....
.....[1]

b. Two parallel metal plates in a vacuum are separated by 0.045m. A potential difference *V* is applied between the plates, as shown in Fig. 5.1.

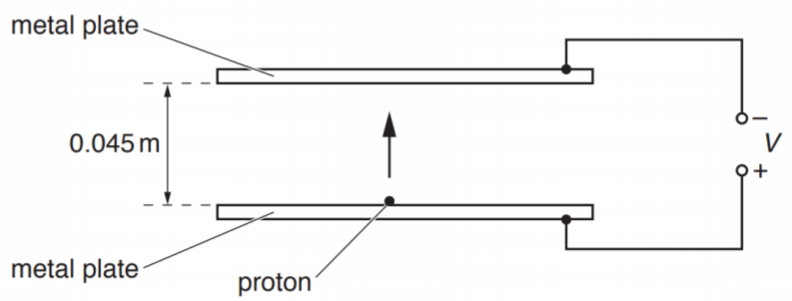

Fig. 5.1

A proton is initially at rest on the surface of the positive plate. The proton in the uniform electric field takes a time of 1.5×10^{-7} s to reach the negative plate.

i. Show that the acceleration of the proton is 4.0×10^{12}ms^{-2}.

[2]

ii. Calculate the electric force on the proton.

force = ... N [1]

iii. Use your answer in **(ii)** to determine

1. the electric field strength,

field strength = ... NC^{-1} [2]

2. the potential difference *V* between the plates.

V = ... V [2]

c. An α particle is now accelerated between the two metal plates in **(b)** by the electric field. Calculate the ratio

$$\frac{\text{acceleration of } \alpha \text{ particle}}{\text{acceleration of proton}}.$$

ratio = ...[2]

[Total: 10]

NO. 2

a. Define *electric field strength*.

.....
.....[1]

b. An electron is accelerated from point A to point B by a uniform electric field, as illustrated in Fig. 3.1.

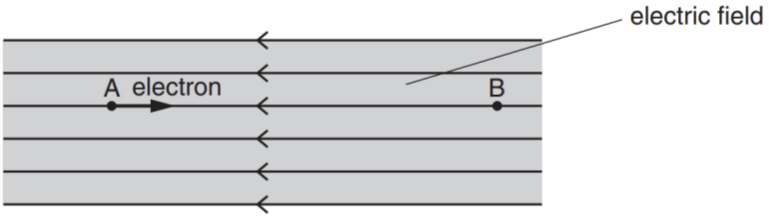

Fig. 3.1

The distance between A and B is 12mm. The velocity of the electron at A is $2.5 kms^{-1}$ and at B is $18 Mms^{-1}$.

Calculate

i. the acceleration of the electron,

acceleration = ...ms^{-2} [2]

ii. the change in kinetic energy of the electron,

change in kinetic energy = ...J [3]

iii. the electric field strength.

electric field strength = ...Vm^{-1} [3]

c. An α-particle moves from A to B in the electric field in **(b)**.
Describe and explain how the change in the kinetic energy of the α-particle compares with that of the electron. Numerical values are not required.

.....
.....
.....
.....
.....
[3]

[Total: 12]

NO. 3

a. i. Define electric potential at a point.

.....
.....
.....[2]

ii. State the relationship between electric potential and electric field strength at a point.

.....
.....
.....[2]

b. Two parallel metal plates A and B are situated a distance 1.2cm apart in a vacuum, as shown in Fig. 6.1.

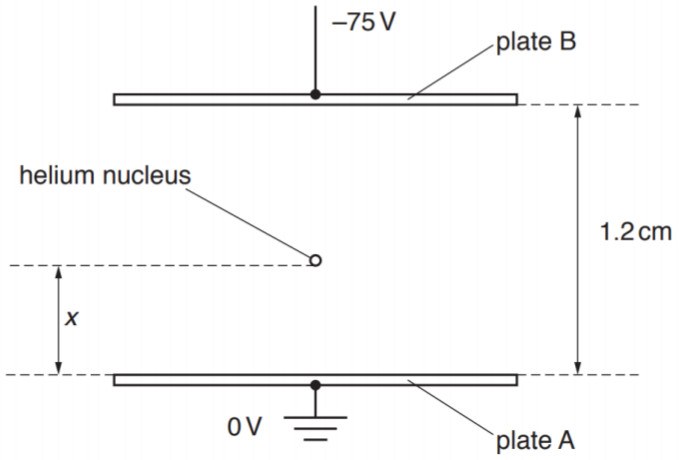

Fig. 6.1

Plate A is earthed and plate B is at a potential of –75V.
A helium nucleus is situated between the plates, a distance x from plate A.
Initially, the helium nucleus is at rest on plate A where $x = 0$.

i. The helium nucleus is free to move between the plates. By considering energy changes of the helium nucleus, explain why the speed at which it reaches plate B is independent of the separation of the plates.

.....
.....
.....
.....[2]

ii. As the helium nucleus ($^{4}_{2}He$) moves from plate A towards plate B, its distance x from plate A increases.
Calculate the speed of the nucleus after it has moved a distance x = 0.40cm from plate A.

speed = ... ms^{-1} [3]

[Total: 9]

NO. 4

a. State the property of an object that experiences a force when the object is placed in:

i. a gravitational field

..... [1]

ii. an electric field.

..... [1]

b. A potential difference of 1.2×10^3V is applied between a pair of horizontal metal plates in a vacuum, as shown in Fig. 3.1.

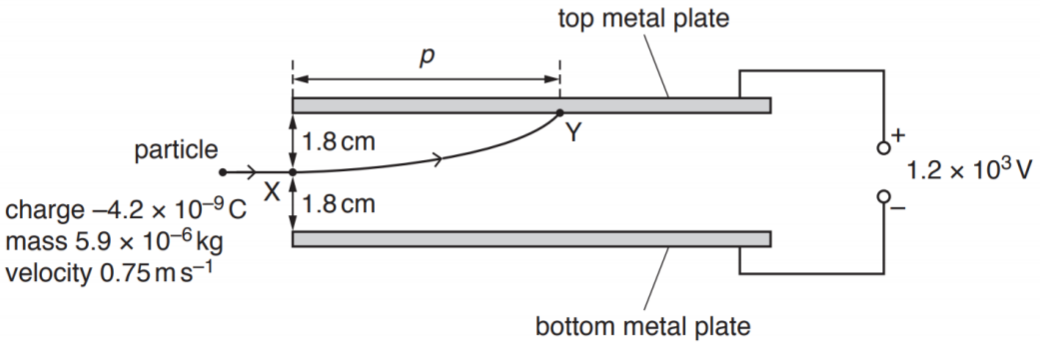

Fig. 3.1 (not to scale)

The separation of the plates is 3.6cm. The electric field between the plates is uniform.

A particle of mass 5.9×10^{-6} kg and charge -4.2×10^{-9}C enters the field at point X with a horizontal velocity of $0.75 ms^{-1}$ along a line midway between the two plates.

The particle is deflected by the field and hits the top plate at point Y.

i. Calculate the magnitude of the electric force acting on the particle in the field.

electric force = ... N [3]

ii. By considering the resultant vertical force acting on the particle, show that the acceleration of the particle in the electric and gravitational fields is $14 ms^{-2}$.

[4]

iii. Determine:

1. the time taken for the particle to move from X to Y

time taken = ... s [2]

2. the distance *p* of point Y from the left-hand edge of the top plate.

p = ...m [1]

[Total: 12]

2024-2025_HW_A2_18. (3)Electric fields_electric field due to two point charges (E and V)_5_50min

NO. 1

a. State an expression for the electric field strength *E* at a distance r from a point charge *Q* in a vacuum. State the name of any other symbol used.

.....
.....
..... [2]

b. Two point charges A and B are situated a distance 10.0cm apart in a vacuum, as illustrated in Fig. 6.1.

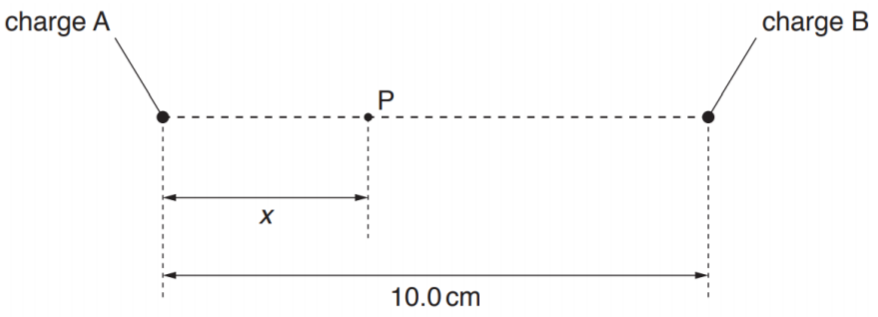

Fig. 6.1

A point P lies on the line joining the charges A and B. Point P is a distance *x* from A.

The variation with distance *x* of the electric field strength *E* at point P is shown in Fig. 6.2.

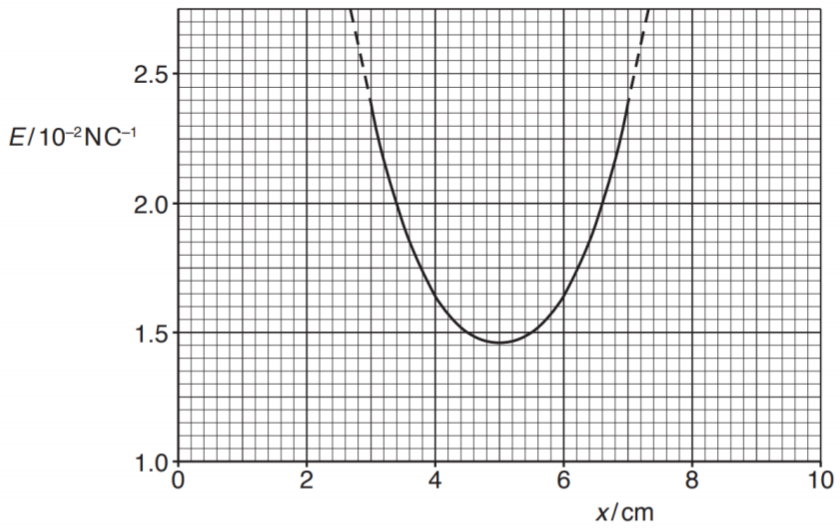

Fig. 6.2

State and explain whether the charges A and B:

i. have the same, or opposite, signs

.....
.....
..... [2]

ii. have the same, or different, magnitudes.

.....
.....
..... [2]

c. An electron is situated at point P.
Without calculation, state and explain the variation in the magnitude of the acceleration of the electron as it moves from the position where x = 3cm to the position where x = 7cm.

.....
.....
.....
.....
.....
..... [4]

[Total: 10]

NO. 2

a. State

i. what is meant by the electric potential at a point,

.....
.....
.....[2]

ii. the relationship between electric potential at a point and electric field strength at the point.

.....
.....
.....[2]

b. Two similar solid metal spheres A and B, each of radius *R*, are situated in a vacuum such that the separation of their centres is *D*, as shown in Fig. 6.1.

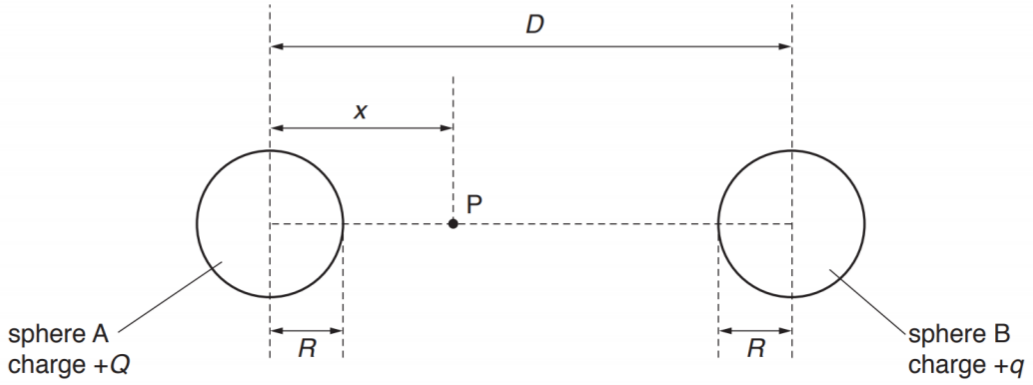

Fig. 6.1

The charge +*Q* on sphere A is larger than the charge +*q* on sphere B.

A movable point P is located on the line joining the centres of the two spheres.

The point P is a distance *x* from the centre of sphere A.

On Fig. 6.2, sketch a graph to show the variation with *x* of the electric potential *V* between the centres of the two spheres.

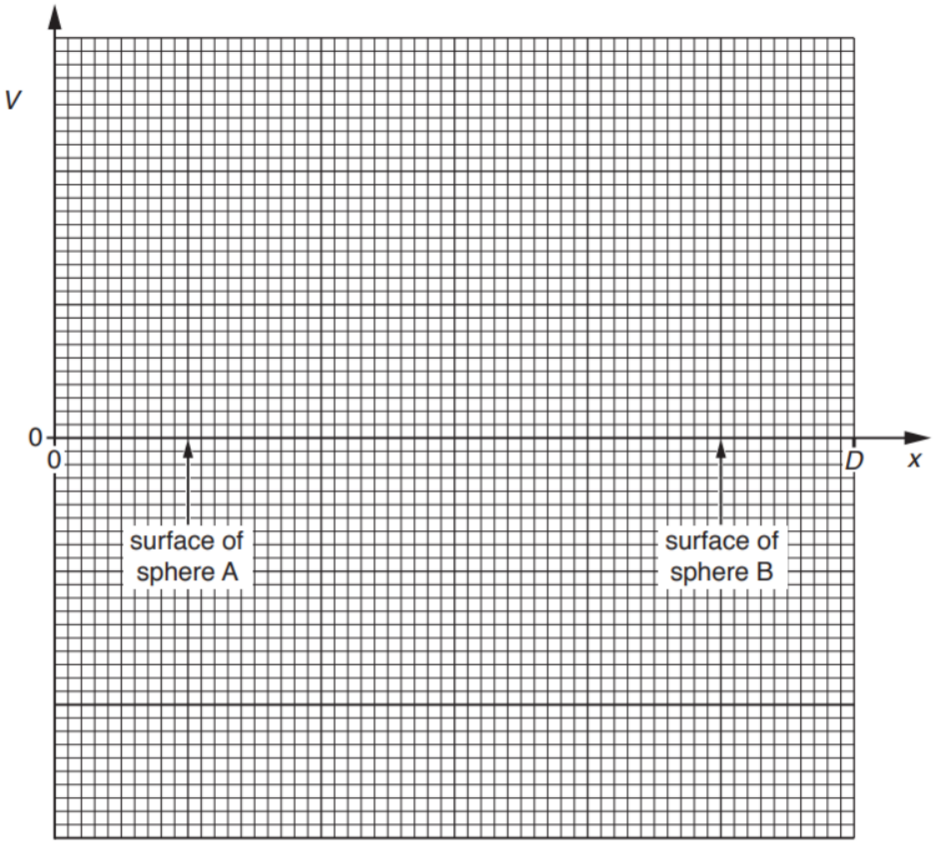

Fig. 6.2

[4]

[Total: 8]

NO. 3

a. Define electric *potential* at a point.

.....
.....
.....[2]

b. Two isolated point charges A and B are separated by a distance of 30.0 cm, as shown in Fig. 4.1.

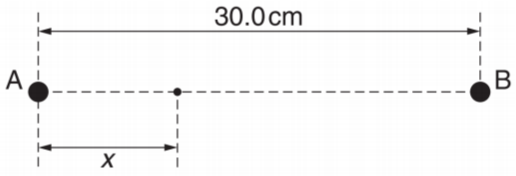

Fig. 4.1

The charge at A is + 3.6 × 10^{-9} C.

The variation with distance x from A along AB of the potential V is shown in Fig.4.2.

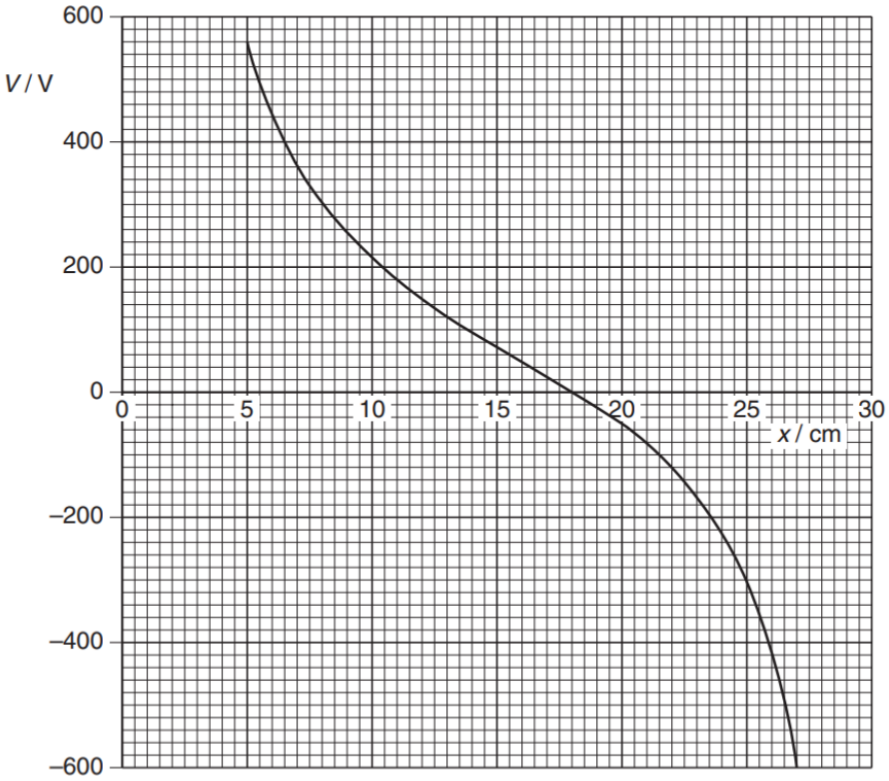

Fig. 4.2

i. State the value of *x* at which the potential is zero.

x = ... cm [1]

ii. Use your answer in (i) to determine the charge at B.

charge = ... C [3]

c. A small test charge is now moved along the line AB in (b) from x = 5.0 cm to x = 27 cm.

State and explain the value of x at which the force on the test charge will be maximum.

.....
.....
.....
.....[3]

[Total: 9]

NO. 4

Two solid metal spheres A and B, each of radius 1.5cm, are situated in a vacuum. Their centres are separated by a distance of 20.0cm, as shown in Fig. 6.1.

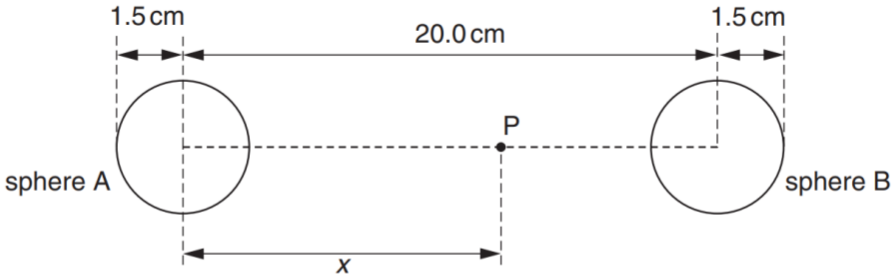

Fig. 6.1 (not to scale)

Both spheres are positively charged.

Point P lies on the line joining the centres of the two spheres, at a distance x from the centre of sphere A.

The variation with distance x of the electric field strength E at point P is shown in Fig. 6.2.

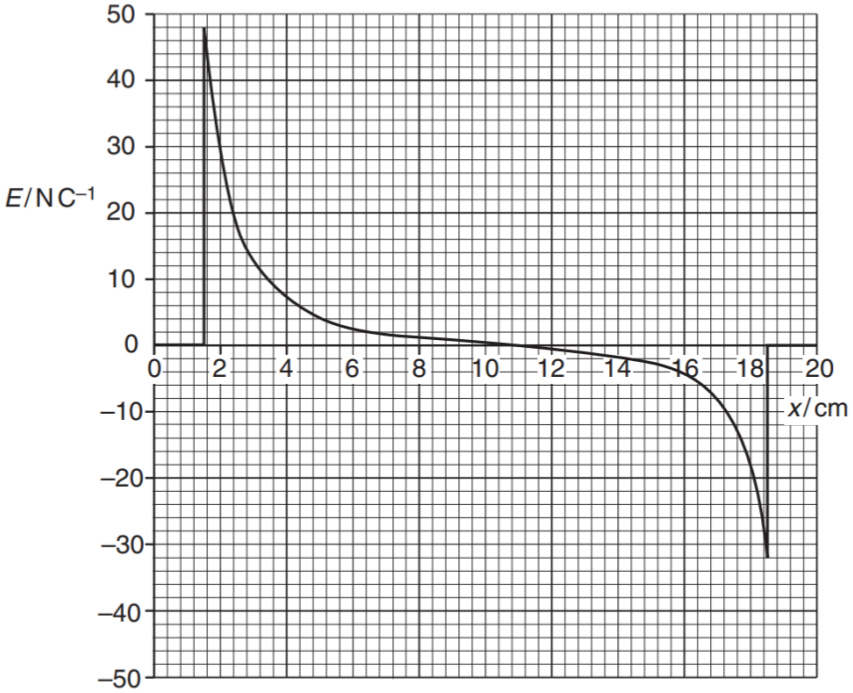

Fig. 6.2

a. Use Fig. 6.2 to determine the ratio

$$\frac{\text{magnitude of charge on sphere A}}{\text{magnitude of charge on sphere B}}.$$

Explain your working.

ratio = ...[3]

b. The variation with distance x of the electric potential V at point P is shown in Fig. 6.3.

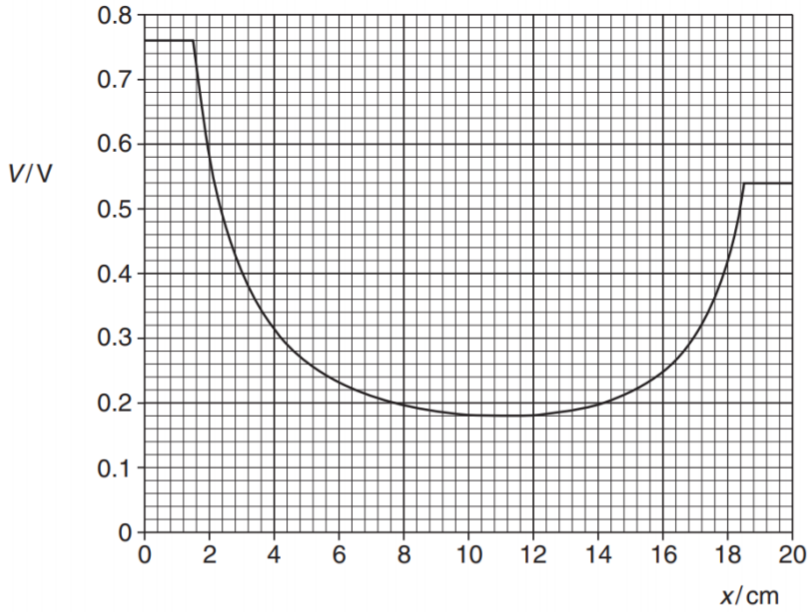

Fig. 6.3

An α-particle is initially at rest on the surface of sphere A.
The α-particle moves along the line joining the centres of the two spheres.

Determine, for the α-particle as it moves between the two spheres,

i. its maximum speed,

maximum speed = ... ms^{-1} [3]

ii. its speed on reaching the surface of sphere B.

speed = ... ms^{-1} [2]

[Total: 8]

NO. 5

Two point charges A and B each have a charge of + 6.4 × 10^{-19} C. They are separated in a vacuum by a distance of 12.0 μm, as shown in Fig. 4.1.

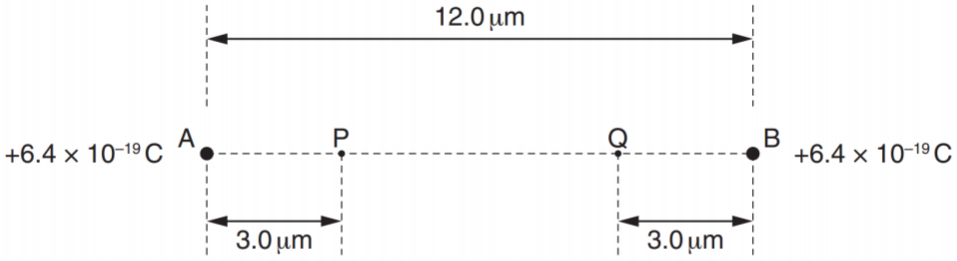

Fig. 4.1

Points P and Q are situated on the line AB. Point P is 3.0 μm from charge A and point Q is 3.0 μm from charge B.

a. Calculate the force of repulsion between the charges A and B.

force = ... N [3]

b. Explain why, without any calculation, when a small test charge is moved from point P to point Q, the net work done is zero.

.....
.....
..... [2]

c. Calculate the work done by an electron in moving from the midpoint of line AB to point P.

work done = ... J [4]